# Reducing Test Anxiety Through Prequestioning: Evidence Across Three Experiments

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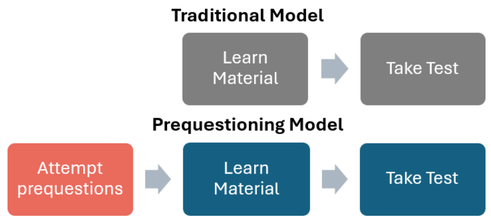
## Background

***Test anxiety*** is regularly associated with poorer learning outcomes such as stress and procrastination[1].

With increasing high-stakes testing, identifying strategies to improve learning and reduce test anxiety are essential[1].

***Prequestioning*** (attempting questions before instruction) improves learning outcomes[2], but its effects on test anxiety remain underexplored.

**The prequestioning effect: testing students *before* learning boosts academic performance[2]**

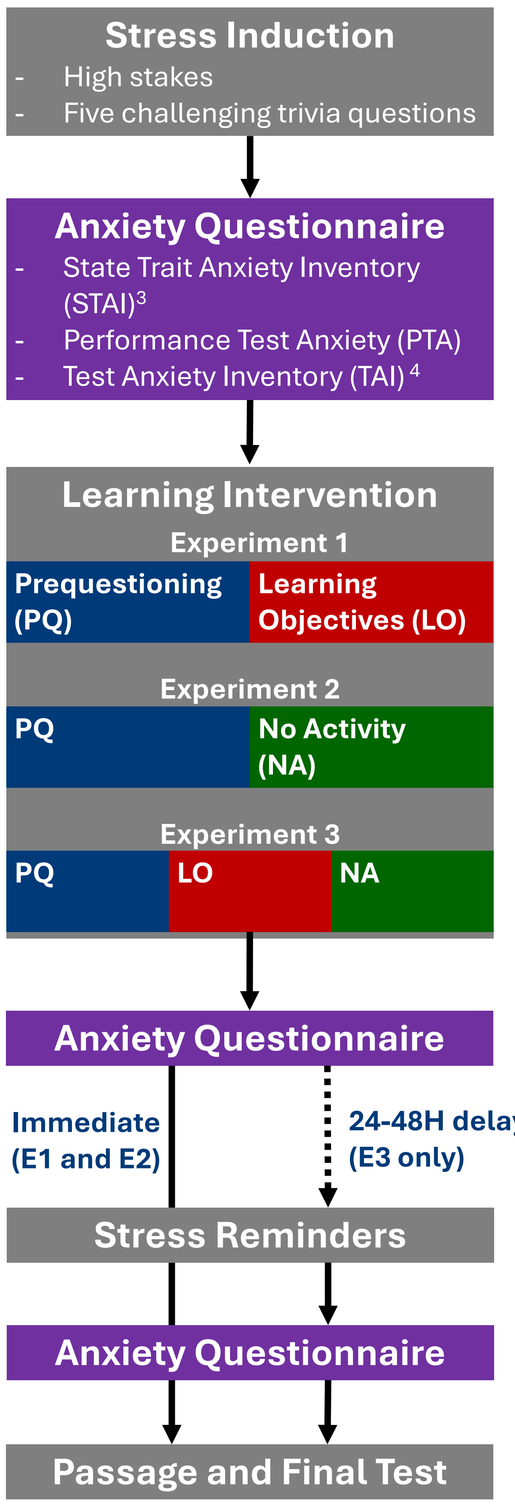



## Hypothesis

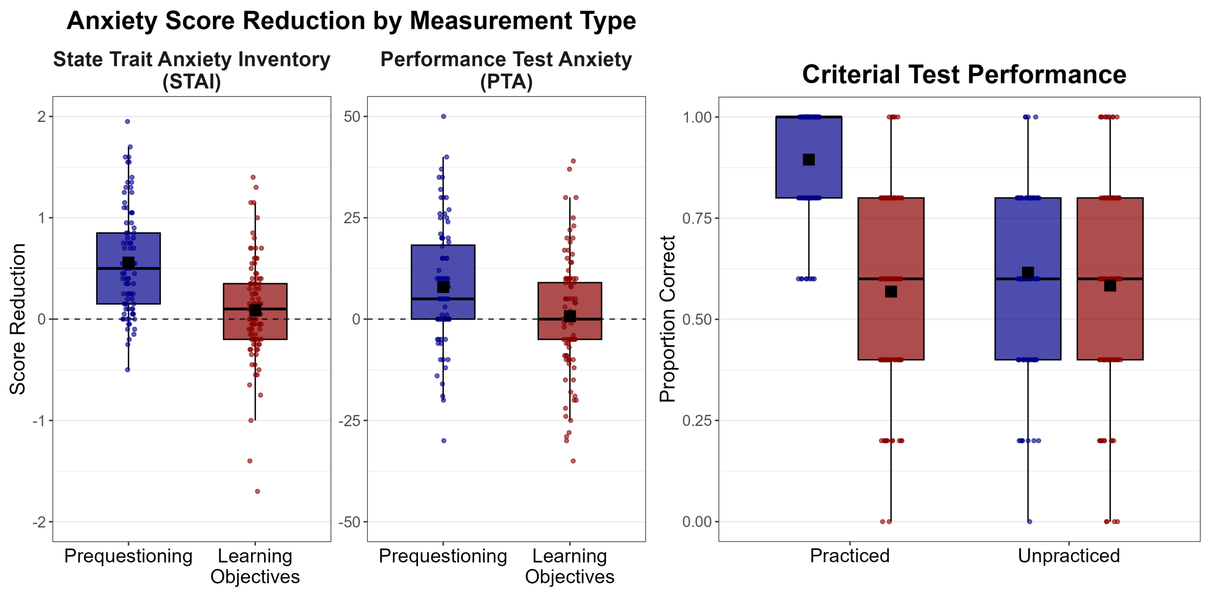
**Across three lab experiments, we examined whether prequestioning reduces test anxiety, and whether any reductions would persist over time.**

**We hypothesized that the prequestioning group would report lower test anxiety levels, and that this effect would persist over a delay.**

## Procedure



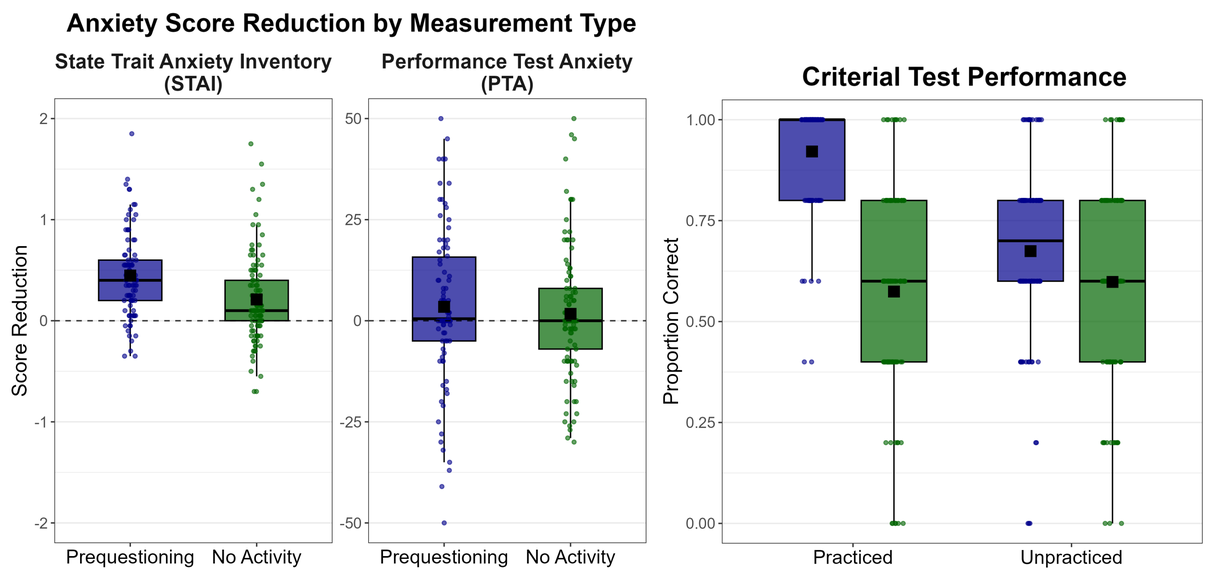

## Experiment 1 Results




**Prequestioning yielded significantly greater anxiety score reductions on both the STAI and PTA compared to the Learning Objectives group.**

**Prequestioning yielded significantly higher scores on the Final Test.**

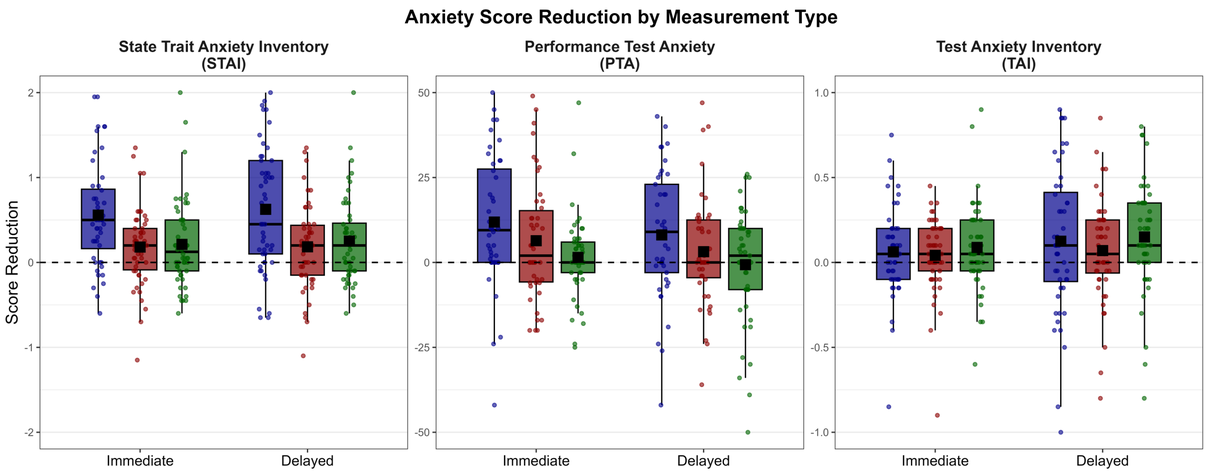
## Experiment 2 Results

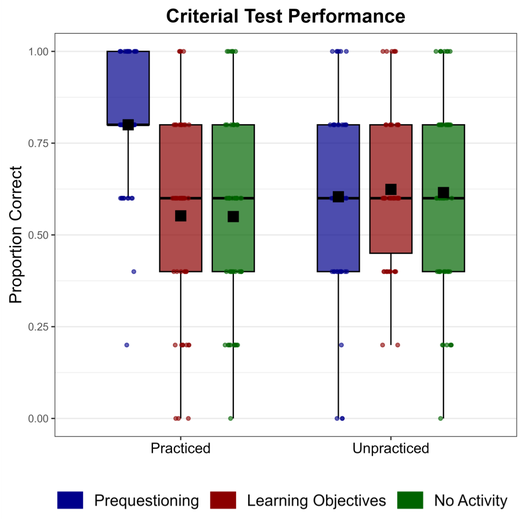



**Prequestioning yielded significantly greater anxiety score reductions on the STAI compared to the No Activity group.**

**Prequestioning yielded significantly higher scores on the Final Test.**

## Experiment 3 Results



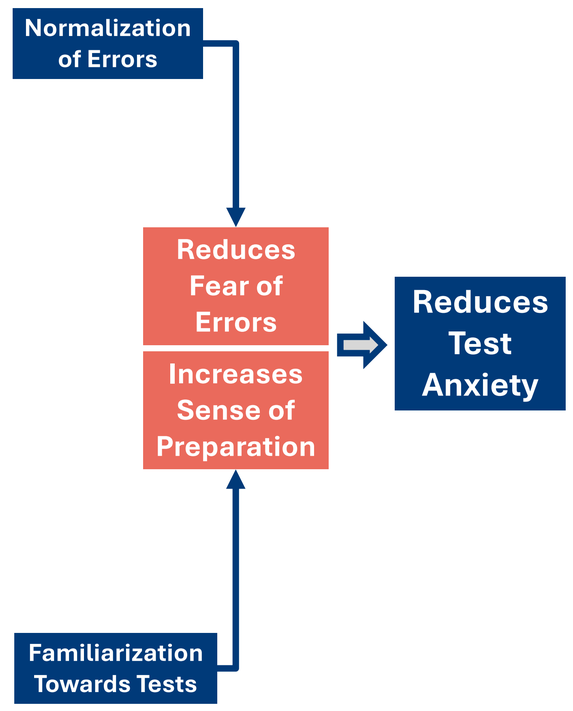




**Prequestioning yielded significantly greater anxiety score reductions on the STAI and PTA compared to the Learning Objectives and No Activity groups.**

**Prequestioning anxiety scores remained low over the 24-48H delay.**

**Prequestioning yielded significantly higher scores on the Final Test.**

## Possible Mechanisms




## Implications and Future Directions

These experiments **provide preliminary evidence that prequestioning reduces test anxiety** compared to learning objectives or no activity, and that these **anxiety-reduction effects last over a delay**.

We encourage educators to utilize prequestioning not only for its learning benefits, but also for its potential to reduce students' test-related anxieties.

Future research will examine prequestioning in classrooms and compare its effectiveness in reducing test anxiety against other learning strategies (e.g., retrieval practice).

## Further Information

**Scan the QR code to access →**
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- Feedback form

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## References

[1]Chakraborty, A. (2023). Exploring the Root Causes of Examination Anxiety: Effective Solutions and Recommendations. *International Journal of Science and Research (IJSR)*, *12*(2), 1096–1102.
[2]Pan, S. C., & Carpenter, S. K. (2023). Prequestioning and Pretesting Effects: A Review of Empirical Research, Theoretical Perspectives, and Implications for Educational Practice. *Educational Psychology Review*, *35*(4), 97.
[3]Spielberger, C. D., Gorsuch, R. L., & Lushene, R. E. (1983). Manual for the State-Trait Anxiety Inventory (STAI). *PaloAlto, CA: Consulting Psychologists Press*.
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